Saurav Kumar Nigam

sauravnigam9752@gmail.com | 9752452555 | Bengaluru | linkedin.com/in/saurav-nigam2002 | github.com/SauravNigam01

Summary

AI/ML Engineer with hands-on experience in **Generative AI**, **Computer Vision**, and end-to-end machine learning solutions. Skilled in building production-ready applications using **LLMs**, **YOLO**, **SQL**, and **Advanced Excel** for data analysis, reporting, and dashboard development. Developed enterprise AI systems during internships, including multimodal compliance solutions, intelligent document extraction pipelines, and AI-powered applications. Strong expertise in **model deployment** and **scalable AI architectures**.

Skills

: Programming: Python, SQL and Java | **Data Analysis & Visualization:** Excel, Power BI and SQL | **Machine Learning:** Supervised Learning, Unsupervised Learning, Classification, Regression, Clustering, Model Training, Model Evaluation, Feature Engineering, Data Preprocessing. | **Deep Learning:** Artificial Neural Networks (ANN), Convolutional Neural Networks (CNN), Recurrent | Neural Networks (RNN), Transformers, Forward and Backpropagation. | **Generative AI:** Large Language Models (LLMs), Prompt Engineering, AI Application Development, | Context-aware AI Systems, LangChain. | **Frameworks & Libraries:** Scikit-learn, TensorFlow, PyTorch, LangChain, NumPy, Pandas, Matplotlib. | **Programming & Tools:** Python, Jupyter Notebook, Google Colab, Git, GitHub

Professional Experience

Qure AI and IIT-BHU: AI/ML Intern
01/2025 – 05/2025 | Varanasi

Developed a **3D brain tumor segmentation model** in collaboration with **Dr. S. K. Singh (IIT -BHU,Varanasi)** using a customized **3D U-Net architecture**, extending 2D models to capture volumetric features from MRI scans.

Designed a deep neural network with **6 encoder and 6 decoder layers**, incorporating **bottleneck layers** and **skip connections** to enhance feature representation and localization accuracy.

Implemented **multi-class segmentation (4 tumor regions)** with advanced preprocessing techniques including **image slicing**, **layering**, and **region truncation** to improve signal-to-noise ratio.

Engineered a **hybrid loss function** combining **Focal Loss**, **Dice Loss**, and **Cross-Entropy Loss** with learnable weighting parameters (α, β) to handle class imbalance and optimize segmentation performance.

Achieved ~**98.6% segmentation accuracy across multiple evaluation metrics**, demonstrating high precision in tumor boundary detection and volumetric consistency.

Optimized training using **3D convolutional networks**, significantly improving spatial coherence over traditional 2D approaches.

Conducted extensive experimentation and validation, and **co-authored a research paper based** on the proposed architecture and results.

Data Science Intern - Innomatics Research
04/2024 – 09/2024 | Bangalore

- Developed an **AI-powered travel planning platform** that generates personalized itineraries based on user preferences, budget, and destination.
- Integrated **real-time weather forecasting using Weather API**, route optimization, **transportation recommendations SERP API**, and cost estimation using **external APIs**.
- Implemented Generative AI capabilities to provide **transportation recommendations**, **trip summaries**, and **conversational assistance**.
- Built an **automated recommendation engine** for hotels, attractions, and transportation options to enhance user experience using **HotelAPI**.
- Designed **scalable backend workflows** for **data processing**, API integration, and dynamic itinerary generation.
- Improved travel decision-making by consolidating weather, route, pricing, and destination insights into a single **AI-driven solution**.

Personal Projects

Credit Card Fraud Detection System | Python, Scikit-learn, FastAPI, Streamlit, Docker | 2025

- Developed an end-to-end machine learning pipeline for credit card fraud detection on **284,807 transactions**, addressing severe class imbalance (0.17% fraud rate) using **SMOTE** oversampling and advanced feature engineering techniques.
- Trained and optimized a Random Forest classifier, achieving ~**99% ROC-AUC** with **high Recall** and **F1-score** for minority class fraud detection, minimizing false negatives in high-risk transactions.
- Built and deployed a real-time fraud prediction system using **FastAPI**, **Streamlit**, and **Docker**, enabling scalable inference with sub-second response latency.
- Implemented model evaluation, performance monitoring, and API-based prediction workflows to support production-ready **machine learning deployment**.

Next Word Predictor | LSTM , NLP , ADAM OPTIMIZER

- Developed a **Next Word Prediction system using Long Short-Term Memory (LSTM) networks** to predict the most probable next word based on input text sequences. Performed end-to-end **NLP preprocessing**,including text cleaning, tokenization, vocabulary generation, and sequence padding, and created training samples using sliding window sequence generation.

Designed a deep learning architecture with **Embedding, LSTM, and Dense layers with Softmax activation**, and trained the model using **categorical cross-entropy loss and Adam optimizer** to improve prediction accuracy and convergence.

Evaluated performance using training and validation metrics and implemented an inference pipeline for **real-time next word prediction**, demonstrating strong understanding of **sequence modeling, language modeling,and deep learning using TensorFlow/Keras and Python**.

Education

Bachelors Of Technology - Samrat Ashok Technological Insitute , Vidisha (M.P)
11/2021 – 07/2025